ABSTRACT

Tick-borne pathogen surveillance in the United States relies predominantly on targeted PCR assays that constrain detection to predefined agents and preclude comprehensive characterization of tick-associated microbiomes. In this study, we report a nanopore metagenomics workflow employing Oxford Nanopore Technologies (ONT) adaptive sampling (NAS) for simultaneous pathogen detection, microbiome profiling, and strain-level genomic characterization in *Ixodes scapularis* ticks identified in Connecticut. When compared to PCR testing, taxonomic classification in our blinded cohorts using our ONT-based approach achieved pathogen-specific precision of 0.909, 0.923, and 0.875 for *Borrelia burgdorferi*, *Babesia microti*, and *Anaplasma phagocytophilum*, respectively, with sensitivity metrics of 0.556, 0.546, and 0.875. Metagenomic profiling additionally revealed known *I. scapularis* endosymbionts, alongside diverse bacterial, fungal, and eukaryotic phyla. Quantitative validation using serially diluted cultured *B. burgdorferi* (strains A3/B31 and C162/297; 0-200 fmol) demonstrated significant dose-dependent read recovery (R^2 = 0.799 and 0.796, respectively; p = 0.0012), uniform coverage, and strain-level discrimination. These results establish ONT metagenomics with NAS as a high-specificity, scalable complement to PCR-based tick surveillance, capable of detecting known pathogens, characterizing microbiome ecology, and providing genomic resolution toward epidemiological strain tracking.